

VENKATESH

Principal SQL Developer | Database Architect & Enterprise Data Engineering Leader

Chicago, IL • (312) 849-6047 • venkatesh.data@dbexcellence.com

linkedin.com/in/ venkatesh-sql-architect • US Permanent Resident

PROFESSIONAL SUMMARY:

Visionary Database Architect and Principal SQL Developer with over 12 years of high-impact experience driving large-scale data platform transformations for Fortune 100 enterprises and high-growth technology firms. Distinguished for architecting the "DataCore 360" enterprise analytics data warehouse, which consolidates 2TB+ of daily transaction data across 40+ source systems and powers mission-critical BI for 3,000+ business users with deterministic sub-3-second query response times. Expert in designing high-performance SQL Server 2022, PostgreSQL, Oracle, and Snowflake database solutions, coupled with advanced T-SQL/PL-SQL optimization, ETL pipeline engineering, and cloud-native AWS/Azure data platform development. Proven mastery in regulated sectors including Fintech (PCI-DSS 4.0), Healthcare (HIPAA), Insurance, and Retail, with a consistent 60% reduction in query execution time and 45% infrastructure cost savings through intelligent database design, indexing strategies, and cloud migration programs. Leadership philosophy centers on "Data Integrity by Design," technical mentorship of high-potential staff, and the relentless pursuit of zero-defect data quality through automated validation frameworks and evidence-based database optimization.

CORE COMPETENCIES:

Database Architecture: Relational Database Design, Dimensional Modeling (Star/Snowflake/Galaxy Schema), Data Vault 2.0, Entity-Relationship Modeling, Normalization (1NF–5NF), OLTP & OLAP Design, Data Lakehouse Architecture

SQL & Stored Procedure Mastery: Advanced T-SQL (SQL Server 2022), PL/SQL (Oracle 19c), PL/pgSQL (PostgreSQL 15), Snowflake SQL, HiveQL, Spark SQL — Window Functions, CTEs, Dynamic SQL, Recursive Queries, Temporal Tables, In-Memory OLTP

ETL & Data Integration: SSIS 2019, AWS Glue, Azure Data Factory, Informatica PowerCenter, Talend, dbt, Apache Airflow, Custom Python ETL pipelines — CDC, SCD Types 1/2/3, Data Lineage, Metadata Management

Performance Engineering: Execution Plan Analysis, Index Strategy (Clustered, Non-Clustered, Columnstore, Filtered, Covering), Query Plan Forcing, Statistics Management, Partition Design, TempDB Optimization, Wait Statistics Analysis

Cloud Data Platforms: AWS (RDS Aurora, Redshift, Glue, Athena, DMS, S3, Lambda), Azure (SQL Managed Instance, Synapse Analytics, Data Factory, SQL Database), Google BigQuery, Snowflake (Enterprise)

High Availability & DR: SQL Server Always On AG, Log Shipping, Database Mirroring, AWS RDS Multi-AZ, Azure SQL Geo-Replication, Point-in-Time Recovery, RPO/RTO Engineering

Data Governance & Security: Row-Level Security (RLS), Column-Level Encryption, Dynamic Data Masking, TDE, AWS KMS, Azure Key Vault, RBAC, HIPAA/HITECH Compliance, PCI-DSS Data Controls, GDPR Anonymization

Engineering Leadership: OKR-Driven Quality Strategy, DBA Team Management (20+ engineers), P&L Responsibility (\$80M+ data platforms), Vendor Management, Cross-functional Stakeholder Alignment, M&A Data Due Diligence

TECHNICAL SKILLS:

Database Platforms & SQL Dialects

Expert Level: SQL Server 2019/2022, PostgreSQL 14/15, Oracle 19c/21c, Snowflake Enterprise, T-SQL, PL/SQL, PL/pgSQL

Advanced Level: MySQL 8.x, Azure SQL Database, AWS Aurora (MySQL/PostgreSQL), Amazon Redshift, Google BigQuery, Azure Synapse Analytics, Spark SQL, HiveQL

Specialty Databases: MongoDB (BSON Aggregation), Cassandra (CQL), DynamoDB, Redis (Sorted Sets/Streams), Elasticsearch, Cosmos DB (SQL/Mongo API), Pinecone (Vector Search)

Advanced T-SQL / SQL Development

Query Engineering: Window Functions (RANK, DENSE_RANK, LAG, LEAD, NTILE, PERCENTILE_CONT), Recursive CTEs, Pivot/Unpivot, Dynamic SQL with sp_executesql, APPLY (CROSS/OUTER), Temporal Tables, JSON/XML parsing

Stored Objects: Stored Procedures, User-Defined Functions (Scalar/Table-Valued/Inline), Views (Indexed/Partitioned), Triggers (AFTER/INSTEAD OF), CLR Integration, Extended Stored Procedures

Performance Objects: Columnstore Indexes (Clustered/Non-Clustered), In-Memory OLTP (Memory-Optimized Tables, Natively Compiled Procedures), Query Store, Resource Governor, Plan Guides

Administrative T-SQL: Dynamic Management Views (DMVs), Wait Statistics Analysis, Index Fragmentation Management, Auto-Statistics Maintenance, TempDB Contention Resolution, Deadlock Graph Analysis

ETL & Data Pipeline Engineering

ETL Tools: SSIS 2019 (Script Tasks, Foreach Loop, Slowly Changing Dimensions), AWS Glue (PySpark), Azure Data Factory (Mapping & Power Query), Informatica PowerCenter 10.x, Talend 7.x

Modern Transformation: dbt Core / dbt Cloud (Incremental Models, Snapshots, Tests, Documentation), Apache Airflow (DAG Orchestration), Prefect, Dagster

Streaming & CDC: Apache Kafka (Kafka Connect JDBC/Debezium), AWS Kinesis, Azure Event Hubs, SQL Server CDC, Oracle GoldenGate, AWS DMS

Cloud Data Platform Engineering

AWS Data Services: RDS Aurora (Multi-AZ, Read Replicas, Global Database), Redshift (RA3 Nodes, Spectrum, Concurrency Scaling), Glue (Catalog, ETL, DataBrew), Athena (Federated Queries), DMS, DataSync, Lake Formation, S3 (Data Lake), Lambda

Azure Data Services: Azure SQL Managed Instance, Azure Synapse Analytics (Dedicated/Serverless Pools), Azure Data Factory, Azure SQL Database (Elastic Pools, Hyperscale), Azure Database for PostgreSQL, Azure Cosmos DB

Snowflake: Virtual Warehouses (Multi-Cluster), Time Travel & Fail-Safe, Zero-Copy Cloning, Snowpipe (Continuous Loading), Dynamic Data Masking, Row Access Policies, Data Sharing

Performance Tuning & DBA

Profiling & Analysis: SQL Server Profiler, Extended Events, Query Store (Forced Plans, Regressed Queries), Execution Plan XML Analysis, Database Engine Tuning Advisor, SolarWinds DPA, SentryOne

Index Strategy: Index Analysis via sys.dm_db_index_usage_stats, Missing Index DMVs, Duplicate Index Detection, Fill Factor Optimization, Filtered Index Design, Partition-Aligned Indexing

HA & DR Tools: SQL Server Always On AG (Synchronous/Asynchronous), Log Shipping, AWS RDS Multi-AZ/Read Replicas, Azure SQL Active Geo-Replication, Ola Hallengren Maintenance Solution

Programming & Scripting

Languages: Python 3.x (pandas, SQLAlchemy, psycopg2, pyodbc), PowerShell (DBATools, SQL Server automation), Bash, Java (JDBC), T-SQL, PL/SQL

DevOps & IaC: Git, GitHub/GitLab, Jenkins, Azure DevOps, Terraform (RDS/Aurora/Redshift provisioning), AWS CDK, Docker (SQL Server containers), Liquibase, Flyway

BI & Reporting

BI Tools: Power BI (DirectQuery/Import, DAX, Dataflows), Tableau (Live/Extract connections), SSRS 2019, SSAS (Tabular/Multidimensional), Amazon QuickSight, Looker, MicroStrategy

Data Governance: Collibra, Apache Atlas, Alation, ERWIN Data Modeler, AWS Lake Formation (Column-level security), Microsoft Purview

PROFESSIONAL EXPERIENCE:

Humana Inc. (Client: CenterWell Pharmacy) | 2022 – Present | Louisville, KY (Remote)

Principal SQL Developer & Database Architect

Strategic Mandate: Recruited as the principal database authority to modernize Humana's pharmacy benefit management and clinical analytics data platform. Lead a team of 18 SQL developers and DBAs across 3 geographic locations, accountable for HIPAA-compliant database architecture covering 25+ SQL Server instances, 4 AWS Aurora clusters, and the enterprise Snowflake data warehouse serving 300+ analytics users.

Key Measurable Achievements:

- **DataCore 360 Warehouse Architecture:** Designed and delivered the enterprise "DataCore 360" Snowflake data warehouse consolidating pharmacy claims (EDI 835/837), member eligibility, clinical encounters, and formulary data from 40+ source systems. Implemented Data Vault 2.0 modeling achieving full historical auditability and zero breaking-change schema evolution for HIPAA compliance.
- **SQL Server 2022 Performance Engineering:** Re-engineered 200+ T-SQL stored procedures supporting pharmacy adjudication and prior authorization workflows, implementing Columnstore indexes, In-Memory OLTP tables for hot-path claim lookups, and Query Store forced plan management, reducing average procedure execution time by 78% and eliminating 95% of blocking/deadlock incidents.
- **AWS Aurora PostgreSQL Migration:** Directed the migration of 8TB of clinical encounter data from legacy SQL Server to AWS Aurora PostgreSQL Multi-AZ with Global Database, designing schema refactoring strategy, Flyway migration scripts, and automated data validation pipelines ensuring zero data loss and <10 minute production cutover window.
- **HIPAA-Compliant Data Masking Framework:** Architected a SQL Server Dynamic Data Masking + Row-Level Security + AWS Lake Formation column-level security framework for PHI data across all environments, enabling analytics access for 300+ users without exposing PII, achieving 100% pass rate on 4 consecutive HIPAA compliance audits.
- **dbt Transformation Layer:** Built a dbt Cloud project on Snowflake implementing 180+ transformation models for HEDIS quality measure calculation, RAF risk adjustment scoring, and population health KPIs, with automated dbt tests (not_null, unique, accepted_values, custom SQL) enforcing data integrity on every pipeline run.
- **ETL Pipeline Modernization:** Migrated 60+ legacy SSIS packages processing pharmacy and clinical data to AWS Glue PySpark jobs orchestrated by Apache Airflow on Amazon MWAA, reducing ETL maintenance overhead by 65% and enabling auto-scaling for peak enrollment period data volumes (10x normal load).
- **Real-Time Claims CDC Pipeline:** Implemented SQL Server CDC + Debezium + Apache Kafka pipeline streaming real-time pharmacy claim adjudication events to Snowflake via Snowpipe, enabling sub-minute data freshness for operational dashboards replacing 4-hour batch loads.
- **Query Performance Program:** Established a Query Performance Center of Excellence, using Extended Events, Query Store regression analysis, and SolarWinds DPA to identify and remediate 50+

top resource-consuming queries quarterly, saving an estimated \$2.1M annually in AWS RDS compute costs.

- **Power BI DirectQuery Optimization:** Redesigned 45 Power BI reports consuming SQL Server and Snowflake directly, implementing aggregation tables, pre-computed summary layers, and indexed views, reducing Power BI report load times from 45 seconds to under 3 seconds for executive dashboards serving the C-suite.

Environment: *SQL Server 2022, T-SQL, Snowflake Enterprise, dbt Cloud, AWS Aurora PostgreSQL, AWS Glue, Apache Airflow (MWAA), Apache Kafka, Debezium, AWS Lake Formation, AWS DMS, Flyway, Python, Power BI, SSRS, Collibra, AWS CDK, Docker, Jenkins*

American Express (Client: Global Commercial Services) | 2019 – 2022 | Phoenix, AZ

Senior SQL Developer & Data Warehouse Engineer

Operational Leadership: Served as the primary SQL and data warehouse authority for American Express Global Commercial Services division. Directed 12 SQL developers building and maintaining PCI-DSS 4.0 compliant data platforms supporting corporate card, expense management, and B2B payment analytics for 65,000+ enterprise clients.

Key Measurable Achievements:

- **Redshift Enterprise Data Warehouse:** Architected a 200TB Amazon Redshift RA3 enterprise data warehouse for commercial card transaction analytics, designing distribution keys, sort keys, and workload management (WLM) queues that reduced complex analytical query execution from 8 minutes to 22 seconds across 5 billion+ transaction records.
- **PCI-DSS T-SQL Security Framework:** Designed a multi-layered PCI-DSS 4.0 compliant SQL Server security architecture implementing Transparent Data Encryption (TDE), Always Encrypted for cardholder data columns, Row-Level Security for agent access partitioning, and automated audit trail generation via SQL Server Audit, passing 3 consecutive PCI-DSS QSA assessments.
- **Oracle to SQL Server Migration:** Led the strategic migration of 300+ Oracle 11g PL/SQL stored procedures, packages, and database jobs to SQL Server 2019 T-SQL using SSMA with custom translation layers for Oracle-specific syntax, delivering zero regression in financial calculation accuracy validated by automated reconciliation test suites.
- **High-Frequency Transaction Optimization:** Re-architected the corporate card authorization database using In-Memory OLTP tables and natively compiled stored procedures for real-time authorization lookups, achieving sub-5ms transaction processing for 12,000+ card authorizations per second while eliminating I/O bottlenecks.
- **Azure Synapse Analytics Implementation:** Delivered Azure Synapse Analytics dedicated SQL pool for commercial spend analytics, implementing Synapse Link from Cosmos DB for real-time operational data integration, designing T-SQL views with column-level security for 500+ business intelligence consumers across finance and risk teams.
- **Automated Index Maintenance Platform:** Built a Python + T-SQL automated index management platform using Ola Hallengren's solution extended with custom DMV-based intelligence, automatically adjusting reorganization vs. rebuild thresholds based on fragmentation levels and maintenance window constraints, reducing index maintenance window by 70%.
- **SOX Compliance Automation:** Developed PowerShell + T-SQL automated SOX compliance evidence generation scripts extracting database permission matrices, change history from SQL Server Audit, and schema change logs from Liquibase, reducing manual audit evidence preparation from 3 weeks to 4 hours per quarterly audit.
- **Snowflake Proof of Concept:** Led an 8-week Snowflake Enterprise POC migrating 50TB of commercial analytics data, demonstrating 85% query performance improvement over Redshift for variable concurrent workloads, providing the business case for a \$12M Snowflake Enterprise licensing investment.

Environment: *SQL Server 2019/2022, T-SQL, Amazon Redshift, Azure Synapse Analytics, Snowflake, Oracle 11g, PL/SQL, SSMA, Azure Data Factory, SSIS 2019, Python, PowerShell, AWS DMS, Liquibase, Tableau, Power BI (DirectQuery), SSAS Tabular, SolarWinds DPA, Jenkins, Terraform*

Target Corporation (Client: Target.com Digital) | 2017 – 2019 | Minneapolis, MN

SQL Developer & Database Performance Engineer

Key Measurable Achievements:

- **Black Friday Database Engineering:** Re-architected SQL Server 2016 order management and inventory databases for Black Friday scale, implementing table partitioning by order date, partition switching for archival, Columnstore indexes for reporting queries, and Read-Scale Always On replicas for BI offloading, handling 250,000 orders/hour at peak with zero downtime.
- **Snowflake Retail Data Mart:** Designed and delivered a Snowflake data mart for Target.com digital analytics, building dbt models for customer 360, product affinity, basket analysis, and promotional lift metrics, enabling 200+ analysts to run interactive queries against 50+ billion clickstream events with sub-10-second response times.
- **ETL Modernization Program:** Migrated 80+ SSIS packages processing product catalog, inventory, and supplier data to Azure Data Factory v2 mapping pipelines with Azure Integration Runtime, implementing tumbling window triggers, retry policies, and Azure Monitor alerting, reducing ETL failure rates from 12% to 0.3%.
- **PostgreSQL Microservices Database Design:** Designed PostgreSQL 11 schemas for 15 Target.com microservices (inventory, pricing, promotions, reviews), implementing PL/pgSQL stored functions, table partitioning for high-volume event tables, and pg_partman for automated partition management supporting 10M+ daily product events.
- **Query Optimization Initiative:** Led a 90-day SQL performance optimization sprint across Target.com's 40 most resource-intensive queries, using Extended Events to profile execution and implementing 180+ index changes, query rewrites, and statistics updates, reducing database server CPU utilization from 87% to 34% during peak load.
- **Automated Database Testing:** Built a Python + pytest-based database testing framework validating stored procedure outputs, ETL transformation logic, and referential integrity constraints automatically on every release branch, catching 35 data integrity regressions before production deployment during the first quarter.

Environment: *SQL Server 2016/2019, T-SQL, Snowflake, PostgreSQL 11, dbt, Azure Data Factory, SSIS 2019, Python, Azure Monitor, Jenkins, Git, Tableau, Power BI, AWS RDS, Liquibase, Docker*

Allstate Insurance Company (Client: Arity Telematics) | 2015 – 2017 | Northbrook, IL

SQL Developer / ETL Engineer

Key Measurable Achievements:

- **Telematics Data Warehouse Design:** Designed and built an AWS Redshift data warehouse consolidating 2B+ daily IoT driving behavior events (acceleration, braking, cornering, phone distraction) from the Drivewise telematics platform, implementing star-schema modeling for actuarial analysis and insurance premium risk scoring.
- **T-SQL Actuarial Calculation Engine:** Developed 120+ complex T-SQL stored procedures and table-valued functions implementing actuarial loss ratio models, premium rating factors, and risk classification algorithms on SQL Server 2014, collaborating with actuarial teams to translate statistical models into optimized database logic.
- **SSIS ETL Pipeline Design:** Built 45 SSIS packages orchestrating nightly data ingestion from 15 insurance source systems including policy administration, claims management, and agency management platforms, implementing SCD Type 2 for policy change history, error logging, and email alerting for pipeline failures.

- **Oracle PL/SQL Optimization:** Optimized 80+ Oracle 12c PL/SQL procedures supporting claims processing and policy servicing operations, rewriting cursor-based logic to set-based operations, adding appropriate indexes, and partitioning large claims history tables, achieving 65% query performance improvement.
- **Power BI Actuarial Dashboards:** Designed SSAS Tabular models and Power BI DirectQuery reports for actuarial loss ratio, combined ratio, and underwriting profitability analysis, enabling actuarial teams to replace Excel-based manual reporting with interactive dashboards refreshed every 4 hours.

Environment: *SQL Server 2014/2016, T-SQL, Oracle 12c, PL/SQL, Amazon Redshift, SSIS 2016, AWS Glue, Python, Power BI, SSAS Tabular, SSRS, Informatica PowerCenter, Jenkins, Git, AWS S3*

Cognizant Technology Solutions (Client: Fidelity Investments) | 2013 – 2015 | Boston, MA

SQL Developer / Database Analyst

Key Measurable Achievements:

- **Mutual Fund Data Warehouse:** Built and maintained SQL Server 2012 dimensional data warehouse for Fidelity's mutual fund performance analytics, designing fact tables for daily NAV, returns, flows, and holdings data with SCD Type 2 dimension management for fund classification, benchmark, and manager change history.
- **T-SQL Performance Tuning:** Identified and resolved 60+ top resource-consuming queries in the fund administration database using SQL Server Profiler and DMVs, implementing missing indexes, rewriting non-SARGable WHERE clauses, and replacing scalar UDFs with inline table-valued functions, reducing batch processing windows from 6 hours to 45 minutes.
- **SSIS Fund Data Pipelines:** Developed 30+ SSIS packages ingesting daily price feeds, corporate action data, and portfolio holding files from Bloomberg, Reuters, and custodian banks into the Fidelity data warehouse, implementing checksum validation, duplicate detection, and automated exception reporting.
- **Stored Procedure Library:** Authored a library of 80+ reusable T-SQL stored procedures, views, and functions for portfolio performance calculation, benchmark comparison, attribution analysis, and regulatory reporting, reducing report development time for BI teams by 55%.
- **Database Migration & Upgrade:** Executed SQL Server 2008 to SQL Server 2012 upgrade for 12 fund administration databases, performing compatibility-level testing, deprecated feature remediation, and post-migration performance validation, achieving zero regression in 500+ stored procedure behaviors.

Environment: *SQL Server 2008/2012, T-SQL, SSIS 2012, Oracle 11g, PL/SQL, Informatica PowerCenter, Tableau, SSRS, SSAS, SQL Server Profiler, Jenkins, Maven, Git*

EDUCATION:

Master of Science in Information Systems & Database Technology – Illinois Institute of Technology (IIT), Chicago, IL

Focus: Database Systems, Data Warehousing, Advanced SQL Optimization, Distributed Data Management | GPA: 3.94/4.0 | Dean's List (All Semesters)

Bachelor of Engineering in Computer Science – PSG College of Technology, Coimbatore, India, 2009–2013

Honors: First Class with Distinction | Best Outgoing Student in Computer Science | President, Database Systems Club